

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI
Second Semester, Session: 2025 – 2026
Control Systems (ECE/EEE/INSTR F242)

MM: 06

Time: 12 Minutes

Quiz - 4

Date 24.03.2026

Note: Write your final answers in the space provided only. Show all the necessary steps in solution.

ID:	Name:	Section:
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1. For unity feedback closed loop system with $G(s)H(s) = K(s+2)/(s(s+1)(s+3))$,
 - (a) Indicate the portion of root locus which will lie on real axis of s-plane. Show it in Fig. 1. [1 M]
 - (b) Calculate the angle of asymptotes and centroid and show it in Fig. 1. [2 M]
 - (c) Find the breakaway point and indicate it clearly in Fig. 1. [2 M]
 - (d) Draw the complete root locus in Fig. 1. [1 M]

(a) $n=3, m=1$; ~~$p_1=0, p_2=1$~~ $p_1=0; p_2=-1; p_3=-3$ & $z_1=-2$
 (Real axis portion shown in Fig. 1)

(b) $\phi_A = \frac{(2q+1)}{(n-m)} \times 180^\circ$; $q=0, 1$
 $\phi_A = 90^\circ, 270^\circ$ ($\because n-m=2$)

$$\sigma_A = \frac{(-1-3) - (-2)}{2} = -1$$

(c). From characteristic equation,
 $K = -\frac{(s^3 + 4s^2 + 3s)}{(s+2)}$; for breakaway point, $dk/ds = 0$.

$$\Rightarrow \frac{dk}{ds} = \frac{d}{ds} \left[\frac{(s^3 + 4s^2 + 3s)}{(s+2)} \right] = 0 \Rightarrow (2s^3 + 10s^2 + 16s + 6) = 0$$

$$\Rightarrow s = -0.534, -2.23 \pm 0.792j$$

↓
valid Breakaway point.

(d).

